Chapter 14 — Atmosphere

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 311–313

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Which gas causes yellow colour in photochemical smog?

(a) CO **(b)** SO_3
(c) NO_2 **(d)** SO_2

Answer: (c) NO_2 — NO_2 is a reddish-brown/yellow gas that colours photochemical smog.

II. Depletion of ozone in the stratosphere is mainly due to reaction of O_3 with:

(a) CO_2 **(b)** SO_2
(c) O_2 **(d)** CFCs

Answer: (d) CFCs — CFCs release chlorine radicals that catalytically destroy ozone.

III. Which particulate matter size is considered the most harmful to human health?

(a) PM 10.0 **(b)** PM 5.0
(c) PM 2.5 **(d)** PM 1.0

Answer: (c) PM 2.5 — PM2.5 penetrates deep into the lungs and bloodstream; it is the standard fine fraction emphasised in air-quality guidelines.

IV. Which is a natural factor that can improve air quality?

(a) Wildfire **(b)** Urbanization
(c) Rainfall **(d)** Industrial emission

Answer: (c) Rainfall — Rainfall washes out particulates and soluble gases (wet deposition).

V. What common human activity increased carbon monoxide in the atmosphere?

(a) Tree planting **(b)** Gasoline driven vehicles
(c) Using electric appliances **(d)** Solar panel installation

Answer: (b) Gasoline driven vehicles — Incomplete combustion of gasoline in vehicles produces CO.

VI. CFCs have been primarily used in:

(a) Fertilizers **(b)** Refrigerants and aerosol propellants
(c) Pharmaceuticals **(d)** Pesticides

Answer: (b) Refrigerants and aerosol propellants — CFCs were used as refrigerants and aerosol propellants.

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VII. What primary chemical process removes CO_2 from the atmosphere?

(a) Combustion **(b)** Photosynthesis
(c) Respiration **(d)** Volcanic activity

Answer: (b) Photosynthesis — Photosynthesis fixes CO_2 into carbohydrates.

VIII. An AQI value between 51 and 100 indicates which level of air quality?

(a) Good **(b)** Moderate
(c) Unhealthy **(d)** Fatal

Answer: (b) Moderate — AQI 51–100 is classified as Moderate.

IX. Acid rain is primarily caused by release of which gases?

(a) CO_2 and CH_4 **(b)** SO_2 and NOx
(c) CFCs and halons **(d)** O_3 and CO

Answer: (b) SO_2 and NOx — SO_2 and NOx form sulfuric and nitric acids in the atmosphere.

X. In sunlight and NOx, VOCs contribute to the formation of:

(a) Acid rain **(b)** Ozone depletion
(c) Photochemical smog **(d)** Greenhouse gases

Answer: (c) Photochemical smog — VOCs + NOx + sunlight form photochemical smog.

XI. Primary pollutants NOx and VOCs react in sunlight to form a key component of photochemical smog, which is a:

(a) Primary pollutant **(b)** Secondary pollutant
(c) Naturally occurring atmospheric gas **(d)** Harmless byproduct of combustion

Answer: (b) Secondary pollutant — Ozone/PAN formed in the atmosphere are secondary pollutants.

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Q.2 — Short Answer Questions

a. Identify and briefly explain three major natural sources of air pollutants.

Volcanic eruptions (SO_2, ash); forest fires (CO, particulates, VOCs); dust storms (particulate matter). Biological decay and pollen also contribute.

b. How can deforestation impact air quality?

Fewer trees means less CO_2 absorption and less filtration of particulates; burning cleared vegetation releases smoke, CO and VOCs, worsening air quality.

c. Explain the reasons for temperature trends in the troposphere and stratosphere.

In the troposphere temperature decreases with altitude because the surface is the main heat source. In the stratosphere temperature increases with altitude because ozone absorbs UV radiation.

d. Describe four anthropogenic activities that deteriorate air quality, naming one pollutant each.

Vehicle exhaust (CO/NOx); industrial combustion (SO_2); construction/demolition (PM10); agriculture/biomass burning (NH_3/particulates).

e. What are the environmental impacts of persistent organic pollutants (POPs)?

POPs resist degradation, bioaccumulate and biomagnify in food chains, and cause toxic/reproductive effects in wildlife and humans.

f. How do polycyclic aromatic hydrocarbons (PAHs) affect human health?

PAHs are carcinogenic and mutagenic; long-term exposure increases cancer risk and can damage the respiratory and immune systems.

g. What is photochemical smog? Under what conditions does it form?

Photochemical smog is a brown, oxidising haze formed when NOx and VOCs react in sunlight. It forms in warm, sunny, stagnant conditions with heavy traffic.

h. What type of data does the air quality index (AQI) system provide?

AQI provides a single number indicating how polluted the air is and the associated health risk, based on concentrations of key pollutants (PM2.5, PM10, O_3, NO_2, SO_2, CO).

i. Distinguish between PM10 and PM2.5, and why PM2.5 is more harmful.

PM10 has diameter ≤ 10 µm; PM2.5 ≤ 2.5 µm. PM2.5 penetrates deeper into the alveoli and can enter the bloodstream, causing more severe respiratory and cardiovascular effects.

j. What are the main chemical processes involved in acid rain formation?

$SO_2 + H_2O \rightarrow H_2SO_3$; $2SO_2 + O_2 \rightarrow 2SO_3$; $SO_3 + H_2O \rightarrow H_2SO_4$. NOx forms HNO_3: $2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$; $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$.

k. What are the specific measures to control smog?

Reduce vehicle emissions (catalytic converters, clean fuels), control industrial NOx/VOC emissions, promote public transport, avoid biomass burning, and enforce air-quality standards.

l. How does a catalytic converter reduce harmful vehicle emissions?

It uses Pt/Pd/Rh to convert CO → CO_2, NOx → N_2, and unburnt hydrocarbons → CO_2 + H_2O before they leave the exhaust.

m. Describe the sources of lead and mercury pollution.

Lead: old leaded petrol, batteries, paints, smelting. Mercury: coal combustion, gold mining, chlor-alkali plants, improper disposal of batteries/fluorescent lamps.

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Descriptive / Long Questions

Q.3 Discuss sources and effects of the following air pollutants: (i) Heavy metals (ii) VOCs (iii) PAHs (iv) POPs.

(i) Heavy metals (Pb, Hg, Cd): from smelting, batteries, coal; cause neurological/kidney damage and bioaccumulate. (ii) VOCs: from solvents, fuels, vegetation; form photochemical smog and some are toxic/carcinogenic. (iii) PAHs: from incomplete combustion; carcinogenic and mutagenic. (iv) POPs: from pesticides/industrial chemicals; persistent, bioaccumulative and toxic over long distances.

Q.4 Write short notes on: (i) CFCs and ozone layer depletion (ii) Greenhouse effect and global warming.

(i) CFCs are stable compounds that reach the stratosphere, where UV releases Cl radicals. $Cl + O_3 \rightarrow ClO + O_2$; $ClO + O \rightarrow Cl + O_2$, destroying ozone catalytically. This thins the ozone layer, increasing UV exposure.

(ii) Greenhouse gases (CO_2, CH_4, H_2O, N_2O) absorb infrared radiation and warm the Earth. Increased concentrations from fossil-fuel burning enhance the greenhouse effect, causing global warming, climate change, sea-level rise and extreme weather.

Q.5 How does fossil fuel burning cause acid rain? Discuss with chemical reactions.

Fossil fuels contain sulfur and produce NOx at high temperature. $S + O_2 \rightarrow SO_2$; $2SO_2 + O_2 \rightarrow 2SO_3$; $SO + H_2O \rightarrow H_2SO_4$. $N_2 + O_2 \rightarrow 2NO$; $2NO + O_2 \rightarrow 2NO_2$; $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$. These acids fall as acid rain, damaging buildings, soils, lakes and vegetation.

Q.6 What is meant by air quality AQI? Describe the factors affecting air quality.

AQI (Air Quality Index) is a numerical scale that communicates how polluted the air is and the associated health effects. It is calculated from concentrations of PM2.5, PM10, O_3, NO_2, SO_2 and CO.

Factors affecting air quality: emissions from vehicles/industry, weather (wind, temperature inversions, rainfall), topography, population density, fuel quality, and seasonal biomass burning.

Quick Check Solutions (in-chapter practice boxes)

Quick Check 14.1 (book p. 299)

a. Mention important air pollutants.

CO, SO_2, NOx, O_3, particulate matter (PM10/PM2.5), VOCs, PAHs, POPs, heavy metals (Pb, Hg) and CFCs.

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b. Equations for formation and depletion of ozone in the stratosphere.

Formation: $O_2 + UV \rightarrow 2O$; $O + O_2 \rightarrow O_3$. Depletion (by CFCs): $CFCl_3 + UV \rightarrow Cl\bullet$; $Cl\bullet + O_3 \rightarrow ClO\bullet + O_2$; $ClO\bullet + O \rightarrow Cl\bullet + O_2$ (net: $O_3 + O \rightarrow 2O_2$).

c. Names and approximate heights of atmospheric layers.

Troposphere (0–12 km), stratosphere (12–50 km), mesosphere (50–85 km), thermosphere (85–600 km), exosphere (above ~600 km).

Quick Check 14.2 (book p. 303)

a. Man-made sources of air pollution.

Vehicle exhaust, industrial emissions, power plants, construction, agriculture, biomass/waste burning, and solvents.

b. How PAHs primarily enter the atmosphere.

Through incomplete combustion of fossil fuels, biomass, tobacco and organic matter (vehicle exhaust, industrial processes, smoking).

c. How VOCs affect air quality.

VOCs react with NOx in sunlight to form photochemical smog and ground-level ozone; some are toxic/carcinogenic.

d. Major sources of heavy metals in the atmosphere.

Smelting, coal combustion, waste incineration, leaded fuel (historically), batteries, and mining/industrial processes.

Quick Check 14.3 (book p. 306)

a. Differentiate classical and photochemical smog.

Classical (London) smog is a reducing mixture of smoke, fog and SO_2, occurring in cool humid conditions. Photochemical (Los Angeles) smog is an oxidising mixture of NOx, VOCs, O_3 and PAN formed in warm sunny conditions.

b. Name greenhouse gases and how they cause global warming.

CO_2, CH_4, N_2O, H_2O vapour, O_3 and CFCs. They absorb infrared radiation emitted by the Earth and re-radiate it, trapping heat and warming the atmosphere.

c. Balanced equation for formation of sulfuric acid from SO_3.

$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$.

d. How HNO_3 is formed from NO in the atmosphere.

$2NO + O_2 \rightarrow 2NO_2$; $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$.

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Quick Check 14.4 (book p. 308)

a. Impact of particulate matter on air quality.

PM reduces visibility, carries toxic substances, and penetrates the respiratory system (PM2.5 deep into lungs/blood), causing respiratory and cardiovascular disease.

b. What is AQI? How does it measure air quality?

The Air Quality Index is a numerical scale that converts concentrations of key pollutants (PM2.5, PM10, O_3, NO_2, SO_2, CO) into a single number indicating pollution level and health risk.

c. Different levels of AQI; safest and most hazardous ranges.

0–50 Good (safest), 51–100 Moderate, 101–150 Unhealthy for sensitive groups, 151–200 Unhealthy, 201–300 Very unhealthy, 301–500 Hazardous (most hazardous).

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